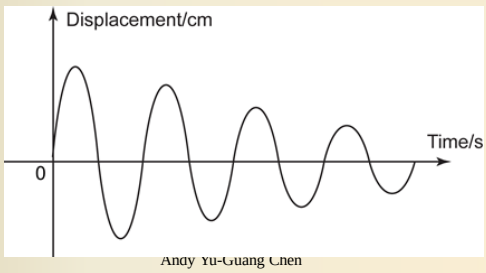
EE1007
Introduction to Computer Science Laboratory

Lab 11
Pointer (II)

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Lab 11.1
Wave Propagation

- **Wave propagation** is any of the ways in which waves travel.
- When the wave travel, the amplitude of the wave will decrease gradually due to the influence of damping.
- An example is shown as follow:



Lab 11.1
Wave Propagation

- In this problem, you are asked to write a program to sort an integer series like wave propagation.
- Your program should take an integer series as the input (the maximum length of the series is 20).
- Your program should sort the input series like wave propagation. The first number of the result will be the largest number in the input series, and the second number of the result will be the smallest one in the input series and so on.
- After sorting, your program should print the result and terminate the program.
- Notice that this program MUST follow the function prototype restriction shown below. Otherwise you won't get any point.
  - **void <function_name>(int *, const int, bool(*)(int, int), bool(*)(int, int));**
    - <function_name> is the name that you named for this function.
- Hint: You may use selection sort to implement this program. (Chapter 7.12)
- the range of each input integer is the same as int type.

Lab 11.1
Wave Propagation

- The screen dialog should appear as follows:
- Example 1 (1 2 3 4 5)

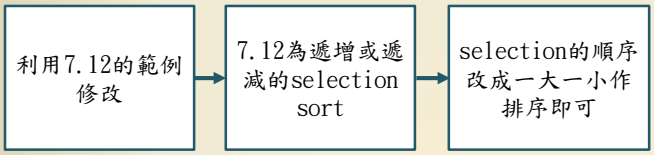
```
1 2 3 4 5
5 1 4 2 3
```

- Example 2 (10 9 8 7 6 5 4 3 2 1)

```
10 9 8 7 6 5 4 3 2 1
10 1 9 2 8 3 7 4 6 5
```

Lab 11.1
Wave Propagation

◆思考方向:



Lab 11.1
Wave Propagation

◆Code:

```
void selectionSort_wave(int *work, const int size, bool(*compare1)(int, int), bool(*compare2)(int, int))
{
    int smallestOrLargest;

    for(int i = 0; i < size - 1; i++)
    {
        smallestOrLargest = i;

        for(int index = i + 1; index < size; index++)
        {
            if(i%2 == 0)
            {
                if(!(*compare1)(*(work+smallestOrLargest), *(work+index)))
                    smallestOrLargest = index;
            }
            else
            {
                if(!(*compare2)(*(work+smallestOrLargest), *(work+index)))
                    smallestOrLargest = index;
            }
        }

        swap(work+i, work+smallestOrLargest);
    }
}
```

Lab 11.1
Wave Propagation

◆Code:

```
void swap(int * const element1Ptr, int * const element2Ptr)
{
    int hold = *element1Ptr;
    *element1Ptr = *element2Ptr;
    *element2Ptr = hold;
}

bool smallest(int a, int b)
{
    return a < b;
}

bool largest(int a, int b)
{
    return a > b;
}
```


Lab 11.2
Matrix Multiplier

- In this problem, you are asked to write a program which can multiply two matrixes.
- The matrix multiplication is shown follow.

$$\begin{bmatrix} a_1 & a_2 & a_3 \\ a_4 & a_5 & a_6 \\ a_7 & a_8 & a_9 \end{bmatrix} \begin{bmatrix} b_1 & b_2 & b_3 \\ b_4 & b_5 & b_6 \\ b_7 & b_8 & b_9 \end{bmatrix} = \begin{bmatrix} c_1 & c_2 & c_3 \\ c_4 & c_5 & c_6 \\ c_7 & c_8 & c_9 \end{bmatrix}$$

- Your program will ask the user to input the number of rows and columns of the two matrixes in the beginning.
- Since the first matrix's column number must equal to the second matrix's row number in a matrix multiplication, your program should always check the size to make sure these two matrixes are able to multiply together.
- If not, your program should print the error message "Cannot multiply these two matrixes." and terminate the execution.

Lab 11.2
Matrix Multiplier

- After that, your program should ask the user to input the first and second matrix based on the size which the user entered before.
- Every elements in the matrix is integer. The matrix will be entered row by row.
- Finally, the program will calculate the result matrix of multiplication and print the result matrix at the end of the program.
- You **MUST** use pointer whenever you try to access any array in the program.
- You are not allowed to use operator [ ] to access any array in the program. If you break this rule, you won't get any points.
- The maximum rows and columns number of a matrix are both 10, and the range of each input integer is $[-2^{63} \sim 2^{63}-1]$.

Lab 11.2
Matrix Multiplier

- The screen dialog should appear as follows:
- Example 1 (can't multiply)

```
Enter rows and columns for first matrix:
3 2
Enter rows and columns for second matrix:
3 2
Cannot multiply these two matrixes.
```

Lab 11.2
Matrix Multiplier

◆ Example 2 (can multiply)

```
Enter rows and columns for first matrix:
2 3
Enter rows and columns for second matrix:
3 2

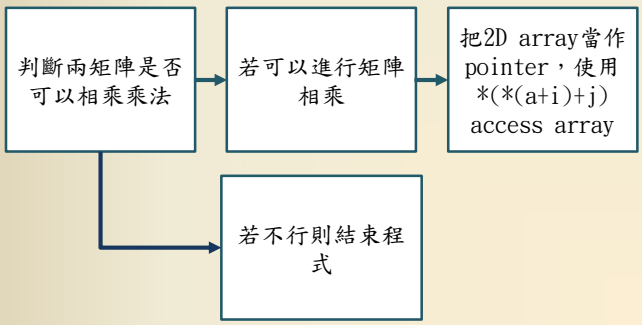
Enter matrix 1:
1 1 1
2 2 2

Enter matrix 2:
1 1
2 2
3 3

Output Matrix:
6 6
12 12
```


Lab 11.2
Matrix Multiplier

◆思考方向:



Lab 11.2
Matrix Multiplier

◆Code:

```
// Storing elements of first matrix.
cout << endl << "Enter matrix 1:" << endl;
for(int i = 0; i < r1; ++i)
    for(int j = 0; j < c1; ++j)
    {
        cin >> *(*(matrix1+i)+j);
    }

// Storing elements of second matrix.
cout << endl << "Enter matrix 2:" << endl;
for(int i = 0; i < r2; ++i)
    for(int j = 0; j < c2; ++j)
    {
        cin >> *(*(matrix2+i)+j);
    }

for(int i = 0; i < r1; ++i)
    for(int j = 0; j < c2; ++j)
    {
        *(*(mult+i)+j) = 0;
    }
```

```
// Multiplying matrix a and b and storing in array mult.
for(int i = 0; i < r1; ++i)
    for(int j = 0; j < c2; ++j)
        for(int k = 0; k < c1; ++k)
        {
            *(*(mult+i)+j) += *(*(matrix1+i)+k) * *(*(matrix2+k)+j);
        }

// Displaying the multiplication of two matrix.
cout << endl << "Output Matrix:" << endl;

for(int i = 0; i < r1; ++i)
    for(int j = 0; j < c2; ++j)
    {
        cout << *(*(mult+i)+j) << " ";
        if(j == c2-1)
            cout << endl;
    }
```


Lab 11.3
NCU Theater Reservation System(III)

- Since more and more students have become a member of the NCU theater, it is getting much more difficult to manage the membership for the theater manager.
- Based on the previous system you had implemented in Lab 10.3, we would like to add some new features in our NCU theater reservation system.
- In this problem, you are asked to develop a function can show all the member's information in the system and integrate it into the NCU theater reservation system.

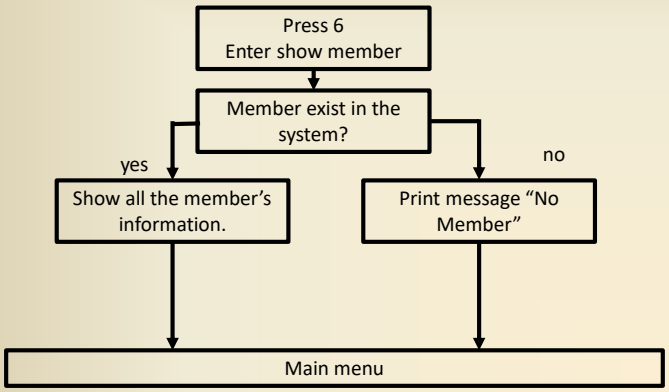
Lab 11.3
NCU Theater Reservation System(III)

◆The new main menu of your system is shown as follows:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here: 5
```


Lab 11.3
NCU Theater Reservation System(III)

◆Dataflow of member system



Lab 11.3
NCU Theater Reservation System(III)

- Once user enters 6, the program will go to show member.
- First, the program will print the message below.
  - "***NCU Theater Membership***"
- Then, the program will check whether a member exist in the system.
- If no, print the message "No Member".
- If yes, show all the member's information, which is username. Notice that the order of the member is according to the creating time. For example, No.1 member is the member who created first in the system.
- After that, return to the main menu.
- The exit function is changed to 7.
- Notice that this function is exactly the same in the member system.

Lab 11.3
NCU Theater Reservation System(III)

- Requirement:
  - If the user enters a wrong input, show the message **"Invalid input, please enter your selection again:"**
  - Notice that the program should work at normal mode when there is no one enter the member system.
  - You **MUST** use the function prototypes shown below to deal with the function of show member, otherwise you will get 0 point.
    - void show_member(…);

Lab 11.3
NCU Theater Reservation System(III)

- The screen dialog should appear as follows:
- Example 1:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here: 5

***NCU Theater Member Center***
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Andy
Enter Password: 0000
Welcome! You are now a member of NCU theater.
Please log in again.
```


Lab 11.3
NCU Theater Reservation System(III)

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here: 6

***NCU Theater Membership***
No.  Username
1.   Andy

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here:
```

Lab 11.3
NCU Theater Reservation System(III)

◆Example 2:

```
*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here: 6

***NCU Theater Membership***
No Member.
```


Lab 11.3
NCU Theater Reservation System(II)

◆Note that there must have one newline between each system.

```
Do you have the membership? (y/n) n
Would you like to sign up? (y/n) y
Enter Username: Andy
Enter Password: 0000
Welcome! You are now a member of NCU theater.
Please log in again.

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here: 6

***NCU Theater Membership***
No.  Username
1.   Andy

*** Welcome to NCU theater reservation system ***
Please type 1 to make a new reservation
Please type 2 to cancel an existing reservation
Please type 3 to show the all available seats
Please type 4 to show the status of all current seats
Please type 5 to member center
Please type 6 to show member
Please type 7 to exit the system
Enter your selection Here:
```

Two blank spaces

Lab 11.3
NCU Theater Reservation System(II)

◆Code:

```
void show_member()
{
    cout << "***NCU Theater Membership***" << endl;
    if(member_data[0][0] == -1)
        cout << "No Member." << endl;
    else
    {
        cout << "No.   " << "Username" << endl;
        for(int i=0; i<member_number; i++)
        {
            string m = member_name[i];
            int temp = i;

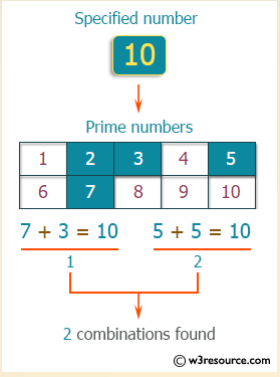
            if(++temp < 10)
              cout << temp << ".    " << m << endl;
            else
              cout << temp << ".   " << m << endl;
        }

    }

    cout << endl;
}
```


Lab 11.4
Goldbach Expectation

- An even number of 4 or more can be represented by the sum of two prime numbers. This is called Goldbach expectation.
- For example, 10 can be expressed as the sum of two prime numbers 7 + 3, 5 + 5.
- You are asked to write a program that accept an integer n from the user and outputs the number of combinations that express n as a sum of two prime numbers.
- The range of n is [4, 50000].
- The execution time restriction of each testcase is 10ms.
- Notice that 7 + 3 and 3 + 7 are considered one combination.
- Hint: you may find all the prime numbers first.



Lab 11.4
Goldbach Expectation

- The screen dialog should appear as follows:
- Example 1 (10)

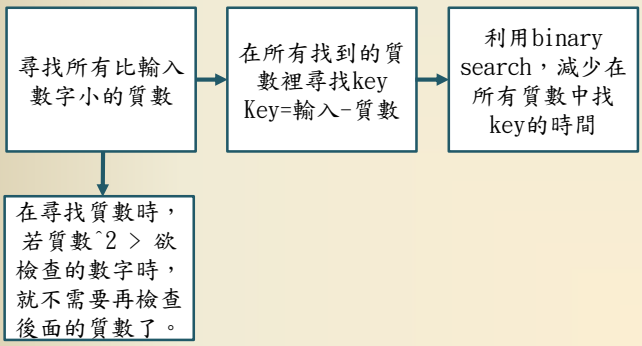
```
Input number: 10
Number of combinations: 2
```

- Example 2 (30)

```
Input number: 30
Number of combinations: 3
```


Lab 11.4
Goldbach Expectation

- 思考方向:



Lab 11.4
Goldbach Expectation

◆Code:

```
#include <iostream>
#include <vector>

using namespace std;


void get_prime(vector<int> *, int);
bool goldbach_expectation(const vector<int> *, int high, int key);

int main()
{

    vector<int> prime;
    prime.push_back(2);
    prime.push_back(3);

    vector<int> *ptr_prime = &prime;

    int n;
    cout << "Input number: ";
    cin >> n;

     get_prime(ptr_prime, n);

    int counter = 0;
    if(n%2 == 0)
    {
        for(int i = 0; prime[i] <= n/2; i++)
        {
            if(goldbach_expectation(ptr_prime, ptr_prime->size()-1, n-prime[i]))
                counter++;
        }
    }
    else
    {
        if(goldbach_expectation(ptr_prime, ptr_prime->size()-1, n-2))
            counter++;
    }

    cout << "Number of combinations: " << counter;

    return 0;
}
```

```
void get_prime(vector<int> *prime, int n)
{

    for(int number = 5; number < n; number++)
    {
        bool t = true;

        for(int i = 0; i < prime->size(); i++)
        {
            if((*prime)[i]*(*prime)[i] > number)
                break;

            else if(number%(*prime)[i] == 0)
            {
                t = false;
                break;
            }
        }

        if(t == true)
            prime->push_back(number);
    }
}

bool goldbach_expectation(const vector<int> *prime, int high, int key)
{
    int low_num = 0;
    int mid_num;

    while(low_num <= high)
    {
        mid_num = (low_num + high)/2;

        if((*prime)[mid_num] == key)
            return true;

        else if((*prime)[mid_num] < key)
            low_num = mid_num + 1;

        else
            high = mid_num - 1;

    }

    return false;
}
```


Evaluation

◆Lab11.1(30%)
- 5 testcases, 6% each

◆Lab11.2(30%)
- 5 testcases, 6% each

◆Lab11.3(30%)
- 5 testcases, 6% each

◆Lab11.4(20%)
- 5 testcases, 4% each